

**EXACT VERIFICATION SUPPLEMENT TO
A COUNTEREXAMPLE TO KOUROVKA NOTEBOOK
PROBLEM 16.45**

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1. SCOPE AND EXPLICIT IDENTIFICATIONS

The paper proves structurally that the explicitly specified group $G = \mathbf{F}_{29}^2 \rtimes H$ has

$$|G| = 100920, \quad b_f(G) = b(G) = 3 < 4 = \mu'(G).$$

Here μ' is the maximum size of an independent subset; b is the maximum size of an inclusion-minimal base over all permutation representations, with bases taken for the permutation image; and b_f restricts this maximum to faithful representations. This supplement describes optional exact checks of the finite inputs and witnesses. These computations are not premises of the structural upper-bound proof. In particular, no subgroup enumeration of the group of order 100920 is asserted.

Over $\mathbf{F}_{29}$ the complement generators are

$$A = \begin{pmatrix} 0 & 28 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 7 \\ 12 & 28 \end{pmatrix}, \quad H = \langle A, B \rangle.$$

The verifiers obtain exactly 120 determinant-one matrices and check $A^2 = B^3 = (AB)^5 = -I$. An explicit isomorphism $\mathrm{SL}_2(5) \rightarrow H$ is specified by

$$\begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix} \mapsto A, \quad \begin{pmatrix} 0 & 4 \\ 1 & 1 \end{pmatrix} \mapsto B.$$

These pairs generate a subgroup of $\mathrm{SL}_2(5) \times H$ of order 120, with both coordinate projections bijective, certifying the isomorphism. The additional checker verifies all $120^2 = 14400$ multiplication identities. Similarly, $A \mapsto (1\ 2)(3\ 4)$ and $B \mapsto (1\ 3\ 5)$ define a quotient onto A_5 with kernel $Z = \{I, -I\}$; all 14400 multiplication identities for this quotient are checked as well.

The independent triple is

$$R_1 = A, \quad R_2 = \begin{pmatrix} 12 & 24 \\ 0 & 17 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 25 & 3 \\ 4 & 4 \end{pmatrix}.$$

The words $R_2 = BAB^2$ and $R_3 = AR_2(AB)^2$, the identities $R_i^2 = -I$, and the product orders $|R_2R_3| = 4$, $|R_1R_3| = 3$, $|R_1R_2| = 10$ are checked. Their omission subgroups have orders 8, 12, 20, intersect in Z , and have core Z in H .

2. COVERAGE OF THE SUBGROUP AND AFFINE CHECKS

The complement subgroups are enumerated by starting with the identity subgroup and, from each discovered subgroup, adjoining every element outside it. Every subgroup has a generating sequence, and each successive prefix occurs along a path in this procedure. The resulting list therefore contains all 76 actual subgroups, not just conjugacy representatives. As a further check, their order distribution is

order	1	2	3	4	5	6	8	10	12	20	24	120
number	1	1	10	15	6	10	5	6	10	6	5	1.

A subgroup family is *meet-irredundant* when removing any member strictly enlarges its total intersection. Each of the $\binom{75}{4} = 1215450$ four-families of proper subgroups is tested; none is meet-irredundant. Since every subfamily of a meet-irredundant family is meet-irredundant, larger ranks are excluded as well. Of the $\binom{75}{3} = 67525$ three-families, 900 are meet-irredundant, but none has trivial intersection. Explicit pairs with trivial intersection give the lower bound for $b_f(H)$; the displayed triple gives the other lower bound. Thus the finite results corroborate $\mu'(H) = 3$ and $b_f(H) = 2$.

All 30 lines of $\mathbf{F}_{29}^2$ are tested, and each stabilizer is cyclic of order 4. The additional checker independently enumerates the 62 subgroups of the scalar group $\mathbf{F}_{29} \rtimes C_4$ and finds no meet-irredundant triple. This is a finite check of the line-subgroup bound used in the proof; the paper supplies the general elementary argument.

Using $(v, h)(w, k) = (v + hw, hk)$, the affine generators

$$(e_1, I), \quad (0, R_1), \quad (0, R_2), \quad (0, R_3)$$

generate 100920 elements. Their omission subgroups have respective orders 120, 6728, 10092, 16820, and every omitted generator is absent from its omission subgroup. The total intersection is $Z_0 = \{(0, I), (0, -I)\}$. Nonnormality is checked directly from

$$(e_1, I)(0, -I)(-e_1, I) = (2e_1, -I) \notin Z_0.$$

The three subgroups

$$\begin{aligned} L_1 &= \langle (e_1, I), (0, -I) \rangle, & L_2 &= \langle (e_2, I), (0, -I) \rangle, \\ L_3 &= \langle (e_1 + e_2, I), (2e_1, -I) \rangle \end{aligned}$$

have order 58. Their pairwise intersections have order 2 and their total intersection is trivial; the additional checker verifies the displayed intersection elements in the paper. Their disjoint coset action has degree $3(100920/58) = 5220$ and a faithful minimal base of size 3.

3. WHY THE CHARACTERISTIC MATTERS

The conclusion does not extend to every characteristic in which such a complement exists. Over $\mathbf{F}_{11}$, put

$$A_{11} = \begin{pmatrix} 0 & 10 \\ 1 & 0 \end{pmatrix}, \quad B_{11} = \begin{pmatrix} 0 & 2 \\ 5 & 1 \end{pmatrix}, \quad C = A_{11}B_{11} = \begin{pmatrix} 6 & 10 \\ 0 & 2 \end{pmatrix}.$$

The first two matrices generate a group of order 120. The element C has order 10, with $C^5 = -I$, and stabilizes the distinct lines $W_1 = \langle(1, 0)\rangle$ and $W_2 = \langle(1, 4)\rangle$; its eigenvalues on them are 6 and 2, respectively. Write $D = \langle C \rangle$, $D_2 = \langle C^5 \rangle$ and $D_5 = \langle C^2 \rangle$. Inside $V \rtimes \langle A_{11}, B_{11} \rangle$, where $V = \mathbf{F}_{11}^2$, take

$$Q_1 = W_1 \rtimes D, \quad Q_2 = W_2 \rtimes D, \quad Q_3 = V \rtimes D_2, \quad Q_4 = V \rtimes D_5.$$

These have orders 110, 110, 242, 605. Since $W_1 \cap W_2 = 0$ and $D_2 \cap D_5 = 1$, their total intersection is 1. The deletion intersections are, in order, W_2, W_1, D_5, D_2 , of orders 11, 11, 5, 2. Every member is therefore essential, giving a faithful minimal base of size 4. This exact witness tests the limit of the argument: the characteristic-29 proof uses $\gcd(120, 28) = 4$ and the resulting cyclic 2-group line stabilizers, whereas $\gcd(120, 10) = 10$ permits this obstruction.

4. REPRODUCTION AND LIMITS

The README gives ordinary Python and C++17 commands and the expected outputs. Python uses only its standard library; all mathematical arithmetic is exact. The reconstruction programs read no precomputed certificate. The two supplied implementations and a newly written checker agree on all 76 subgroups as sets of actual matrices. The two Python implementations additionally agree on the digest of the complete sorted affine element set. Saved expected outputs are used only for subsequent comparison; elapsed runtime is not a mathematical assertion.

These programs check explicit finite claims and provide reproducibility; they do not replace the structural proof of the universal bounds, establish bibliographic novelty, or imply external human peer review. The paper and checks were prepared with AI assistance. No proof-assistant formalization is claimed.